PROBLEM SET 7

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 50) **(Extra Credit Max Score:** 15, on a 100-point scale)

1) **(10 pts)** Compute the matrix elements of X and P in the $|n\rangle$ basis and show it is the same as the result from Exercise 7.3.4.
hint: start from $X_{n'n} = \langle n'|X|n\rangle$, and $P_{n'n} = \langle n'|P|n\rangle$, and re-write X and P in terms of the raising and lowering operators.

Exercise 7.4.1 (p. 212) (5 pts.)

$$\langle n'|\hat{X}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle n'|\hat{a}^{+} + \hat{a}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n+1}\langle n'|n+1\rangle + \sqrt{n}\langle n'|n-1\rangle\right)$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n+1}\,\delta_{n';n+1} + \sqrt{n}\,\delta_{n';n-1}\right) \checkmark$$

$$\langle n'|\hat{P}|n\rangle = i\sqrt{\frac{m\omega\hbar}{2}}\langle n'|\hat{a}^{+} - \hat{a}|n\rangle = i\sqrt{\frac{m\omega\hbar}{2}}\left(\sqrt{n+1}\,\delta_{n';n+1} - \sqrt{n}\,\delta_{n';n-1}\right) \checkmark$$

2) **(10 pts)** Find the $\langle X\rangle, \langle P\rangle, \langle X^2\rangle, \langle P^2\rangle, \Delta X \cdot \Delta P$ in the state $|n\rangle$
hint: for the expectation values $\langle X\rangle = \langle n|X|n\rangle$, similarly for $\langle P\rangle$, you can start from the general expression derived in problem (1) and set $n = n'$ in that case.

Exercise 7.4.2 (p.212) (10pts.)

$$\hat{X} = \sqrt{\frac{\hbar}{2m\omega}}\,(\hat{a}^+ + \hat{a}) \qquad \hat{P} = i\sqrt{\frac{m\hbar\omega}{2}}\,(\hat{a}^+ - \hat{a})$$

From class

$$\langle n|\hat{X}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n+1}\,\delta_{n,n+1} + \sqrt{n}\,\delta_{n,n-1}\right) = 0$$

$$\langle n|\hat{P}|n\rangle = i\sqrt{\frac{m\hbar\omega}{2}}\left(\sqrt{n+1}\,\delta_{n,n+1} - \sqrt{n}\,\delta_{n,n-1}\right) = 0$$

Next

$$\langle n|\hat{X}^2|n\rangle = \frac{\hbar}{2m\omega}\langle n|(\hat{a}^+ + \hat{a})^2|n\rangle = \frac{\hbar}{2m\omega}\langle n|\underbrace{(\hat{a}^+)^2}_{\sim\delta_{n,n+2}=0} + \hat{a}^+\hat{a} + \hat{a}\hat{a}^+ + \underbrace{\hat{a}^2}_{\sim\delta_{n,n-2}=0}|n\rangle$$

$$= \frac{\hbar}{2m\omega}\left(\langle n|\hat{a}^+\hat{a}|n\rangle + \langle n|\hat{a}\hat{a}^+|n\rangle\right)$$

$$= \frac{\hbar}{2m\omega}\left(\langle n|\hat{a}^+\sqrt{n}|n-1\rangle + \langle n|\hat{a}\sqrt{n+1}|n+1\rangle\right)$$

$$= \frac{\hbar}{2m\omega}\left(\sqrt{n}\cdot\sqrt{n}\langle n|n\rangle + \sqrt{n+1}\sqrt{n+1}\langle n|n\rangle\right) = \frac{(2n+1)\hbar}{2m\omega}$$

$$\langle n|\hat{P}^2|n\rangle = -\frac{m\hbar\omega}{2}\langle n|(\hat{a}^+ - \hat{a})^2|n\rangle = -\frac{m\hbar\omega}{2}\langle n|\underbrace{\hat{a}^{+2}}_{0} - \hat{a}^+\hat{a} - \hat{a}\hat{a}^+ + \underbrace{\hat{a}^2}_{0}|n\rangle$$

$$= +\frac{m\hbar\omega}{2}\langle n|\hat{a}^+\hat{a} + \hat{a}\hat{a}^+|n\rangle = (2n+1)\frac{m\hbar\omega}{2}$$

$$\Rightarrow \quad \Delta P \cdot \Delta X = \sqrt{\left(\langle\hat{X}^2\rangle - \underbrace{\langle\hat{X}\rangle^2}_{0}\right)\left(\langle\hat{P}^2\rangle - \underbrace{\langle\hat{P}\rangle^2}_{0}\right)}$$

$$= (2n+1)\sqrt{\frac{\hbar}{2m\omega}\cdot\frac{m\hbar\omega}{2}} = \left(n + \frac{1}{2}\right)\hbar$$

3) **(10 pts)** The virial theorem in classical mechanics states that for a particle bound by a potential $V(r) = ar^k$, the average (over the orbit) kinetic and potential energies are related by $\bar{T} = c(k)\bar{V}$, where $c(k)$ depends only on k.

(a) **(5 pts)** Show that $c(k) = k/2$ by considering a circular orbit.

7.4.3

For a circular orbit:

$$F = \frac{\partial V}{\partial r} = kar^{k-1}$$

$$F = mv^2/r$$

$$\Rightarrow mv^2/r = kar^{k-1}$$

$$\Rightarrow mv^2 = kar^k$$

$$\Rightarrow T = \frac{1}{2}mv^2 = \frac{1}{2}kar^2 = \frac{k}{2}V.$$

Therefore, $c(k) = \frac{1}{2}k$.

(b) **(5 pts)** Using results from the previous exercise (2) show that for the oscillator ($k = 2$) $\langle T \rangle = \langle V \rangle$ in the quantum state $|n\rangle$

For the quantum oscillator:

$$\langle T \rangle = \frac{1}{2m}\langle p^2 \rangle$$

$$\langle p^2 \rangle = -\frac{m\omega\hbar}{2}\langle n|(a^+ - a)^2|n\rangle$$

$$= -\frac{m\omega\hbar}{2}\langle n|a^+a^+ - a^+a - aa^+ + aa|n\rangle$$

$$\langle n|a^+a^+|n\rangle = \sqrt{n}\cdot\sqrt{n+1}\ \langle n-1|n+1\rangle = 0$$

$$\langle n|a^+a|n\rangle = \sqrt{n}\cdot\sqrt{n}\ \langle n-1|n-1\rangle = n$$

$$\langle n|aa^+|n\rangle = \sqrt{n+1}\cdot\sqrt{n+1}\ \langle n+1|n+1\rangle = n+1$$

$$\langle n|aa|n\rangle = \sqrt{n+1}\cdot\sqrt{n}\ \langle n+1|n-1\rangle = 0$$

$$\Rightarrow \langle p^2 \rangle = \frac{m\omega\hbar}{2}(n + n + 1) = \frac{m\omega\hbar^2}{2}(2n+1)$$

$$\Rightarrow \langle T \rangle = \frac{\hbar\omega}{4}(2n+1)$$

$$\langle V \rangle = \tfrac{1}{2} m\omega^2 \langle X^2 \rangle$$

$$\langle X^2 \rangle = \frac{\hbar}{2m\omega} \langle n|(a+a^+)^2|n\rangle$$

$$= \frac{\hbar}{2m\omega} \langle n|\, aa + aa^+ + a^+a + a^+a^+ |n\rangle$$

$$= \frac{\hbar}{2m\omega} (n + n + 1)$$

$$= \frac{\hbar}{2m\omega} (2n+1)$$

$$\Rightarrow \langle T \rangle = \frac{\hbar\omega}{4} (2n+1)$$

Therefore, $\langle T \rangle = \langle V \rangle$.

4) **(5 pts)** Show that $\langle n|X^4|n\rangle = (\hbar/2m\omega)^2[3 + 6n(n+1)]$
hint: remember to first write X in terms of the $(a, a^\dagger)$ operators, as was done in the example covered in class

7.4.4

$$\langle n|x^4|n\rangle = \langle n| \frac{\Delta^4}{4}(a + a^\dagger)^4 |n\rangle$$

$$= \frac{1}{4}\left(\frac{\hbar}{m\omega}\right)^2 \langle n| [\, aaa^\dagger a^\dagger + aa^\dagger aa^\dagger + a^\dagger aaa^\dagger + a^\dagger aa^\dagger a + a^\dagger a^\dagger aa + aa^\dagger a^\dagger a \,]|n\rangle$$

(the other terms do not have equal numbers of a's and $a^\dagger$'s)

$$= \frac{1}{4}\left(\frac{\hbar}{m\omega}\right)^2 \langle n|[\, 6\, aaa^\dagger a^\dagger - (1+2+3+4+2)\, aa^\dagger + (1+2) \,]|n\rangle$$

since $\quad a^\dagger a = a a^\dagger - 1$

$$a a^\dagger a a^\dagger = a a a^\dagger a^\dagger - a a^\dagger$$

$$a^\dagger a a a^\dagger = a a a^\dagger a^\dagger - 2 a a^\dagger$$

$$a^\dagger a a^\dagger a = a a a^\dagger a^\dagger - a^\dagger a - 2 a a^\dagger$$

$$= a a a^\dagger a^\dagger - 3 a a^\dagger + 1 \quad , \text{ etc} \ldots$$

$$= \frac{1}{4}\left(\frac{\hbar}{m\omega}\right)^2 \left[6\,(n+1)(n+2) - 12\,(n+1) + 3 \right]$$

since $\quad |n\rangle = \dfrac{1}{\sqrt{n!}}\, a^{\dagger\, n}\, |0\rangle$

$$a^\dagger |n\rangle = \frac{1}{\sqrt{n!}}\, a^{\dagger\, n+1} |0\rangle = \sqrt{n+1}\; |n+1\rangle$$

$$\langle n | a a^\dagger | n\rangle = n+1$$

$$a^\dagger a^\dagger |n\rangle = \sqrt{(n+1)(n+2)}\; |n+2\rangle$$

$$\langle n | a a a^\dagger a^\dagger | n\rangle = (n+1)(n+2)$$

Thus

$$\langle n | x^4 | n\rangle = \frac{1}{4}\left(\frac{\hbar}{m\omega}\right)^2 \left[6(n+1)(n+2-2) + 3 \right]$$

$$= \frac{3}{2}\left(\frac{\hbar}{m\omega}\right)^2 \left[n(n+1) + \frac{1}{2} \right]$$

5) **(15 pts)** Consider the three angular momentum variables in classical mechanics:

$$l_x = yp_z - zp_y$$
$$l_y = zp_x - xp_z$$
$$l_z = xp_y - yp_x$$

(a) **(5 pts)** Construct L_x, L_y, and L_z, the quantum counterparts in terms of the position (X, Y, Z) and momentum (P_x, P_y, P_z) operators and note that there are no ordering ambiguities. **hint:** don't overthink it

7.4.8

(1) $$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y$$
$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z$$
$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$$

(b) **(5 pts)** Verify that $\{l_x, l_y\} = l_z$ [see Eq. (2.7.3) or class notes on Ch. 10 for the definition of Poisson Brackets]

(2) $$\{l_x, l_y\} = \{yp_z - zp_y,\ zp_x - xp_z\}$$
$$= \left(\frac{\partial l_x}{\partial x}\frac{\partial l_y}{\partial p_x} - \frac{\partial l_x}{\partial p_x}\frac{\partial l_y}{\partial x}\right) + \left(\frac{\partial l_x}{\partial y}\frac{\partial l_y}{\partial p_y} - \frac{\partial l_x}{\partial p_y}\frac{\partial l_y}{\partial y}\right)$$
$$+ \left(\frac{\partial l_x}{\partial z}\frac{\partial l_y}{\partial p_z} - \frac{\partial l_x}{\partial p_z}\frac{\partial l_y}{\partial z}\right)$$

$$= (0-0) + (0-0) + (p_y x - y p_x)$$
$$= x p_y - y p_x$$
$$= l_z.$$

(c) **(5 pts)** Verify that $[L_x, L_y] = i\hbar L_z$

hint: start from your results in part (a); work with the position/ momentum operators, **NO** need to substitute $P_i \rightarrow -i\hbar\partial/\partial_i,\ X_i \rightarrow x_i$

(3) $$\begin{aligned}[\hat{L}_x, \hat{L}_y] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y,\ \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] \\ &= [\hat{y}\hat{p}_z,\ \hat{z}\hat{p}_x] - [\hat{y}\hat{p}_z,\ \hat{x}\hat{p}_z] - [\hat{z}\hat{p}_y, \hat{z}\hat{p}_x] + [\hat{z}\hat{p}_y,\ \hat{x}\hat{p}_z] \\ &= \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x - 0 - 0 + \hat{x}[\hat{z}, \hat{p}_z]\hat{p}_y \\ &= -i\hbar\hat{y}\hat{p}_x + i\hbar\hat{x}\hat{p}_y \\ &= i\hbar(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x) \\ &= i\hbar\hat{L}_z.\end{aligned}$$

Extra Credit

(Points obtained here will be added to Final)

6) **(Extra Credit)** **(15 pts)** *Thermodynamics of Oscillators*. The Boltzmann formula

$P(i) = e^{-\beta E(i)}/Z$, where $Z = \sum_i e^{\beta E(i)}$, gives the probability of finding a system in a state i with energy $E(i)$, when it is in thermal equilibrium with a reservoir of absolute temperature $T = 1/\beta k$, where $k = 1.4 \times 10^{-16}$ ergs/°K is Boltzmann's constant.

(a) **(3 pts)** Show that the thermal average of the system's energy is

$$\bar{E} = \sum_i E(i)P(i) = -\frac{\partial}{\partial \beta} \ln Z$$

Problem 4: Exercise 7.5.4

(1) $\bar{E} \equiv \sum_i E(i)\, P(i) = \sum_i E(i) e^{-\beta E(i)}/Z = (1/Z) \sum_i E(i) e^{-\beta E(i)} = -(1/Z) \sum_i (d/d\beta)(e^{-\beta E(i)}) = -(1/Z)(d/d\beta)(\sum_i e^{-\beta E(i)}) = -(1/Z)(dZ/d\beta) = -d(\ln Z)/d\beta.$

(b) **(3 pts)** Let the system be a classical oscillator. The index i is now continuous and corresponds to the variables x and p describing the states of the oscillator, i.e.,

$$i \to x, p$$

$$\sum_i \to \iint dx dp$$

$$E(i) \to E(x,p) = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2$$

Show that the classical partition function is

$$Z_{\text{cl}} = \sqrt{\frac{2\pi}{\beta m \omega^2}}\sqrt{\frac{2\pi m}{\beta}} = \frac{2\pi}{\omega\beta}$$

and that the classical average energy is given by

$$\bar{E}_{\text{cl}} = \frac{1}{\beta} = kT \text{ (note that } \bar{E}_{\text{cl}} \text{ is independent of } m \text{ and } \omega\text{).}$$

(2) $Z_{\text{cl}} = \sum_i e^{-\beta E(i)}$ becomes, upon replacing the sum over states by an integral over phase space,

$$\begin{aligned} Z_{\text{cl}} &= \int dx dp\, e^{-\beta E(x,p)} = \int dx dp \exp\left\{-\beta\left(\frac{p^2}{2m} + \frac{m\omega^2 x^2}{2}\right)\right\} \\ &= \left(\int_{-\infty}^{\infty} dx\, e^{-\beta m\omega^2 x^2/2}\right)\left(\int_{-\infty}^{\infty} dp\, e^{-\beta p^2/(2m)}\right) = \left(\frac{2\pi}{\beta m\omega^2}\right)^{1/2}\left(\frac{2\pi m}{\beta}\right)^{1/2} = \frac{2\pi}{\omega\beta}. \end{aligned}$$

So, $\bar{E}_{\text{cl}} = -d(\ln Z_{\text{cl}})/d\beta = -(d/d\beta)\ln(2\pi/\omega\beta) = (d/d\beta)[\ln\beta - \ln(2\pi/\omega)] = 1/\beta$.

(c) **(3 pts)** For the quantum oscillator the quantum number n plays the role of the index i. Show that the quantum partition function is

$$Z_{\text{qu}} = e^{-\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega})^{-1}$$

and that the quantum average energy is given by

$$\bar{E}_{\text{qu}} = \hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1}\right)$$

(3) $Z_{\text{qu}} = \sum_i e^{-\beta E(i)}$ becomes, upon replacing the sum by a sum over the energy eigenbasis,

$$Z_{\text{qu}} = \sum_n e^{-\beta E_n} = \sum_{n=0}^{\infty} e^{-\beta(n+\frac{1}{2})\hbar\omega} = e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} \left(e^{-\beta\hbar\omega}\right)^n = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}.$$

Thus

$$\begin{aligned}
\bar{E}_{\text{qu}} &= -\frac{d \ln Z_{\text{qu}}}{d\beta} = -\frac{1}{Z_{\text{qu}}} \frac{dZ_{\text{qu}}}{d\beta} = \frac{e^{-\beta\hbar\omega} - 1}{e^{-\beta\hbar\omega/2}} \frac{d}{d\beta} \left(\frac{e^{-\beta\hbar\omega/2}}{e^{-\beta\hbar\omega} - 1} \right) \\
&= \frac{e^{-\beta\hbar\omega} - 1}{e^{-\beta\hbar\omega/2}} \left(\frac{-(\hbar\omega/2)e^{-\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega}) - e^{-\beta\hbar\omega/2}(\hbar\omega)e^{-\beta\hbar\omega}}{(1 - e^{-\beta\hbar\omega})^2} \right) \\
&= \hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1} \right).
\end{aligned}$$

(d) **(3 pts)** It is intuitively clear that as the temperature T increases (and $\beta = 1/kT$ decreases) the oscillator will get more and more excited and eventually (from the correspondence principle)

$\bar{E}_{\text{qu}} \rightarrow \bar{E}_{\text{cl}}$ as $T \rightarrow \infty$

Verify that this is indeed true and show that "large T" means $T \gg \hbar\omega/k$.

(4) From the above expression

$$\lim_{T\to\infty} \bar{E}_{\text{qu}} = \lim_{\beta\to 0} \bar{E}_{\text{qu}} = \hbar\omega \left(\frac{1}{2} + \frac{1}{\lim_{\beta\to 0}(e^{\beta\hbar\omega} - 1)} \right).$$

If $\beta\hbar\omega \ll 1$, then $e^{\beta\hbar\omega} \approx 1 + \beta\hbar\omega + \mathcal{O}(\beta\hbar\omega)^2$, so $\lim_{\beta\to 0}(e^{\beta\hbar\omega} - 1) \approx \beta\hbar\omega$ and therefore

$$\lim_{T\to\infty} \bar{E}_{\text{qu}} = \lim_{\beta\to 0} \hbar\omega \left(\frac{1}{2} + \frac{1}{\beta\hbar\omega} \right) \approx \frac{1}{\beta} = \bar{E}_{\text{cl}}.$$

(e) **(3 pts)** Consider a crystal with N_0 atoms, which for small oscillations, is equivalent to $3N_0$ decoupled oscillators. The mean thermal energy of the crystal $\bar{E}_{\text{crystal}}$ is $\bar{E}_{\text{cl}}$ or $\bar{E}_{\text{qu}}$ summed over all the normal modes. Show that if the oscillators are treated classically, the specific heat per atom is $C_{\text{cl}}(T) = \dfrac{1}{N_0}\dfrac{\partial \bar{E}_{\text{crystal}}}{\partial T} = 3k$ which is independent of T and the parameters of the oscillators and hence the same for crystals. This agrees with experiment at high temperatures but not as $T \to 0$. Empirically, $C(T) \to 3k$ (T large) and $C(T) \to 0$ ($T \to 0$).

Following Einstein, treat the oscillators quantum mechanically, assuming for simplicity that they all have the same frequency ω.

Show that $C_{\text{qu}}(T) = 3k\left(\dfrac{\theta_E}{T}\right)^2 \dfrac{e^{\theta_E/T}}{(e^{\theta_E/T}-1)^2}$

where $\theta_E = \hbar\omega/k$ is called the *Einstein temperature* and varies from crystal to crystal. Show that

$C_{\text{qu}}(T) \to 3k$ as $T \gg \theta_E$

$C_{\text{qu}}(T) \to 3k\left(\dfrac{\theta_E}{T}\right)^2 e^{-\theta_E/T}$ as $T \ll \theta_E$

Although $C_{\text{qu}}(T) \to 0$ as $T \to 0$, the exponential falloff disagrees with the observed $C(T) \to T^3$ behavior as $T \to 0$. This discrepancy arises from assuming that the frequencies of all normal modes are equal, which is of course not generally true. This discrepancy was eliminated by Debye, but Einstein's simple picture by itself is remarkably successful (see Fig. 7.3)

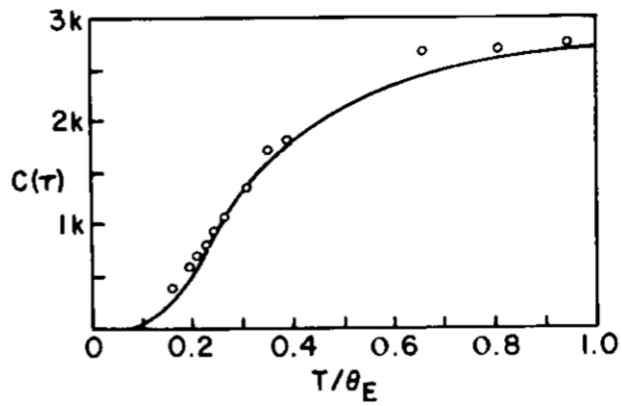

Figure 7.3. Comparison of experiment with Einstein's theory for the specific heat in the case of diamond. (θ_E is chosen to be 1320 K.)

(5) Using the result from part (2) that for each of the $3N_0$ modes (which I'll label by an index $i = 1, \ldots, 3N_0$) $\bar{E}_{\rm cl}(i) = \beta^{-1} = kT$,

$$C_{\rm cl} \equiv \frac{1}{N_0}\frac{d\bar{E}_{\rm cl}}{dT} = \frac{1}{N_0}\frac{d}{dT}\sum_{i=1}^{3N_0} \bar{E}_{\rm cl}(i) = \frac{1}{N_0}\frac{d}{dT}\sum_{i=1}^{3N_0} kT = \frac{1}{N_0} 3N_0 k = 3k.$$

Quantum mechanically, using the result from part (3) for each mode gives

$$\begin{aligned} C_{\rm qu} &\equiv \frac{1}{N_0}\frac{d\bar{E}_{\rm qu}}{dT} = \frac{1}{N_0}\frac{d}{dT}\sum_{i=1}^{3N_0} \hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1}\right) \\ &= \frac{3N_0\hbar\omega}{N_0}\frac{d}{dT}\left(\frac{1}{2} + \frac{1}{e^{\theta_E/T} - 1}\right) = \frac{-3\hbar\omega}{(e^{\theta_E/T} - 1)^2}\frac{d}{dT}(e^{\theta_E/T} - 1) \\ &= \frac{-3\hbar\omega}{(e^{\theta_E/T} - 1)^2}\left(\frac{-\theta_E}{T^2}\right)e^{\theta_E/T} = 3k\left(\frac{\theta_E}{T}\right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T} - 1)^2}, \end{aligned}$$

where in the second line we have written $\beta\hbar\omega = \hbar\omega/(kT) \equiv \theta_E/T$.
For $T \gg \theta_E$, $e^{\theta_E/T} \approx 1 + (\theta_E/T) + \cdots$, so

$$C_{\rm qu} \approx 3k\left(\frac{\theta_E}{T}\right)^2 \frac{1 + \frac{\theta_E}{T} + \cdots}{(\frac{\theta_E}{T} + \cdots)^2} \approx 3k.$$

For $T \ll \theta_E$, $e^{\theta_E/T} \gg 1$, so $e^{\theta_E/T} - 1 \approx e^{\theta_E/T}$ and

$$C_{\rm qu} \approx 3k\left(\frac{\theta_E}{T}\right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T})^2} = 3k\left(\frac{\theta_E}{T}\right)^2 e^{-\theta_E/T}.$$